

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>

Concise solution :

- Use de Broglie with the electron rest energy : $\lambda = h / (m_{ec} \sqrt{1 + (2E/c^2)})$, giving $\lambda (m) = 527 (eV/c^2)^{-1/2} \lambda (Dm)$ for electron masses .
- Convert 1 eV to momentum : $E = \frac{1}{2} c^2 = (1 eV / 1.602 \times 10^{-19} J/eV) \approx 6.22 \times 10^{-34}$ erg s . With $c = 2.998 \times 10^{10}$ cm/s , $\frac{1}{2} = 9.109 \times 10^{-31}$ kg , $h = 6.625 \times 10^{-27}$ erg Å·s .
- Compute the square root : $\sqrt{1 + (2E/c^2)} \approx 6.21 \times 10^3$ erg Å·s .
- Compute wavelength : $\lambda \approx (6.625 \times 10^{-27}) / (9.109 \times 10^{-31} \times 6.21 \times 10^3) \approx 1.23 \times 10^{-7}$ cm .

Correct answer: G. 1.23×10^{-7} cm

</think>

<answer>

G. 1.23×10^{-7} cm

</answer>

